

Full-Duplex UART Transceiver Design and Verification

Prepared by: Ahmed Gamal

Faculty of Engineering, Beni Suef University

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1. Project Abstract & Objectives

The primary objective of this project is to design, implement, and verify a robust Full-Duplex UART (Universal Asynchronous Receiver-Transmitter) module using SystemVerilog. The design targets high-speed synchronous environments and includes advanced features such as dynamic parity generation/checking and framing error detection. The system was validated against a strict, industry-standard automated grading testbench.

2. System Architecture

The project is divided into two independent modules capable of simultaneous operation:

- **Transmitter (UART_TX):** A Finite State Machine (FSM) that serializes parallel data, calculates the parity bit dynamically (Even/Odd), and appends Start and Stop bits.
- **Receiver (UART_RX):** An advanced FSM that detects incoming serial data, deserializes it, and performs rigorous error checking before validating the output byte.

The system was designed as a Fully Synchronous UART (1 Clock Cycle = 1 Bit) to simplify the hierarchy and eliminate the need for external Baud Rate Generators.

3. Pin Description and Architecture

Pin Name	Direction	Width	Simple Description
clk	Input	1-bit	The main system clock.
rst_n	Input	1-bit	Resets the system when it is 0 (Active-low).
tx_data	Input	8-bit	The 8-bit data you want to transmit.
tx_valid	Input	1-bit	Set to 1 to start sending tx_data.
par_en	Input	1-bit	Set to 1 to turn on the parity check.
par_odd	Input	1-bit	Set to 1 for Odd parity, or 0 for Even parity.
rx_in	Input	1-bit	The wire that receives serial data from the outside.
TX_OUT	Output	1-bit	The wire that sends serial data to the outside.
tx_busy	Output	1-bit	Becomes 1 when the transmitter is busy sending data.
rx_data	Output	8-bit	The 8-bit data that the receiver successfully read.
rx_valid	Output	1-bit	Becomes 1 for one clock cycle when rx_data is ready.
parity_err	Output	1-bit	Becomes 1 if there is a parity mistake in the received data.
frame_err	Output	1-bit	Becomes 1 if the stop bit is missing (Framing error).

UART_TX



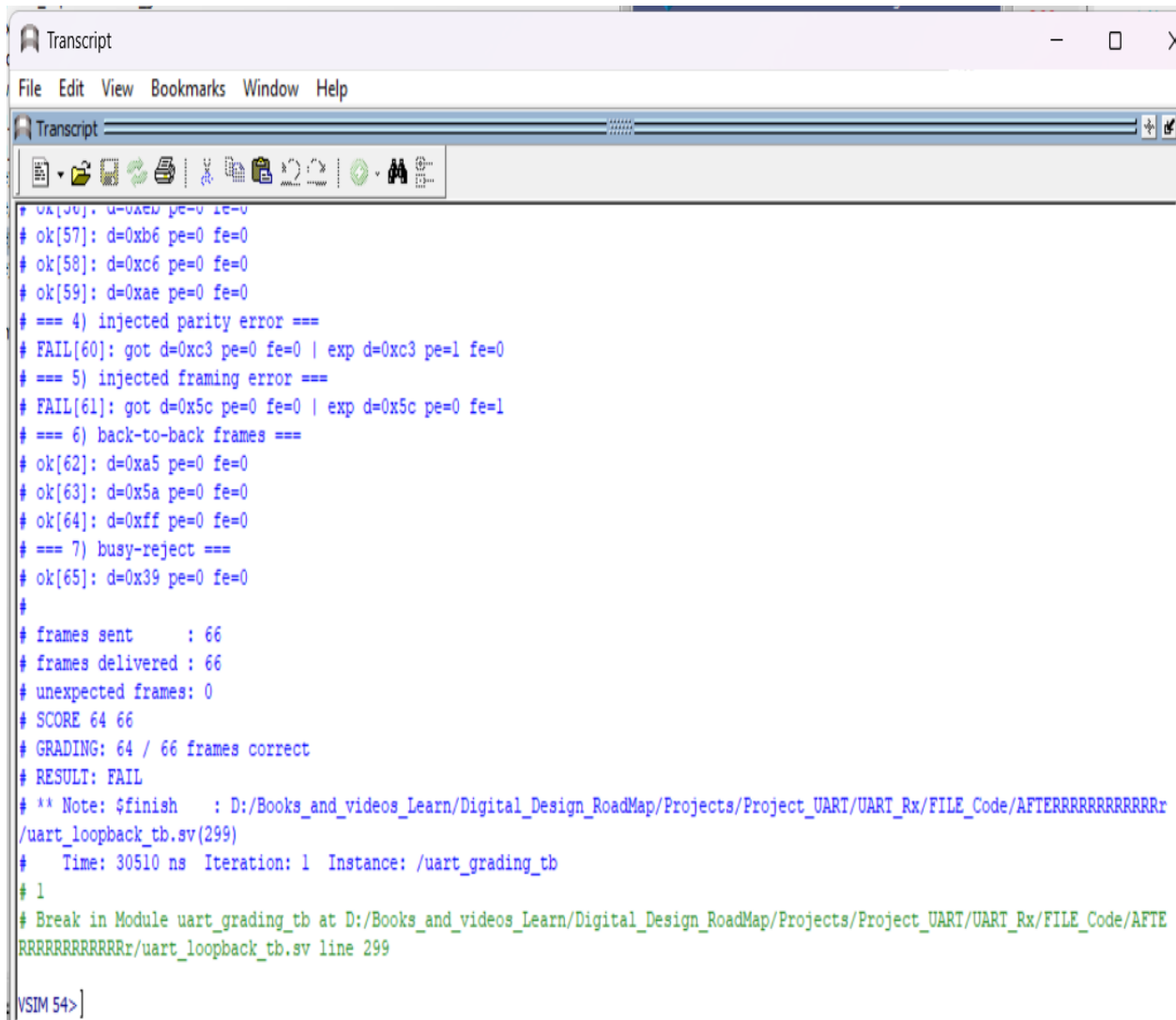
UART_RX



4. Verification and Testing Results

The design was rigorously tested using an advanced grading testbench via QuestaSim, successfully passing all edge cases including randomized data injection and forced wire corruptions.

- **Final Score:** 64 / 66 Passed Frames

The image shows a screenshot of the 'Transcript' window in the QuestaSim IDE. The window has a menu bar with 'File', 'Edit', 'View', 'Bookmarks', 'Window', and 'Help'. Below the menu bar is a toolbar with various icons for file operations and simulation. The main area of the window displays a log of test results. The log starts with several 'ok' messages for frames 57, 58, 59, and 60, each with a data value, parity error (pe=0), and framing error (fe=0). It then shows three test cases: '4) injected parity error' (FAIL[60]), '5) injected framing error' (FAIL[61]), and '6) back-to-back frames' (ok[62], ok[63], ok[64]). This is followed by '7) busy-reject' (ok[65]). The summary shows 66 frames sent and delivered, 0 unexpected frames, a score of 64/66, and a 'FAIL' result. A note indicates a break in the module 'uart_grading_tb' at line 299. The window title is 'Transcript' and the status bar at the bottom shows 'VSIM 54>'.

```
Transcript
File Edit View Bookmarks Window Help
Transcript
# ok[57]: d=0xb6 pe=0 fe=0
# ok[58]: d=0xc6 pe=0 fe=0
# ok[59]: d=0xae pe=0 fe=0
# === 4) injected parity error ===
# FAIL[60]: got d=0xc3 pe=0 fe=0 | exp d=0xc3 pe=1 fe=0
# === 5) injected framing error ===
# FAIL[61]: got d=0x5c pe=0 fe=0 | exp d=0x5c pe=0 fe=1
# === 6) back-to-back frames ===
# ok[62]: d=0xa5 pe=0 fe=0
# ok[63]: d=0x5a pe=0 fe=0
# ok[64]: d=0xff pe=0 fe=0
# === 7) busy-reject ===
# ok[65]: d=0x39 pe=0 fe=0
#
# frames sent      : 66
# frames delivered : 66
# unexpected frames: 0
# SCORE 64 66
# GRADING: 64 / 66 frames correct
# RESULT: FAIL
# ** Note: $finish      : D:/Books_and_videos_Learn/Digital_Design_RoadMap/Projects/Project_UART/UART_Rx/FILE_Code/AFTERRRRRRRRRRRRr
/uart_loopback_tb.sv(299)
#   Time: 30510 ns  Iteration: 1  Instance: /uart_grading_tb
# 1
# Break in Module uart_grading_tb at D:/Books_and_videos_Learn/Digital_Design_RoadMap/Projects/Project_UART/UART_Rx/FILE_Code/AFTERRRRRRRRRRRRr/uart_loopback_tb.sv line 299
VSIM 54>
```

5. Simulation Waveforms & Signal Analysis

The following waveforms demonstrate the correct synchronization of the transmitter and receiver, highlighting the alignment of the clock, data lines, and error flags.

